Cantera vs NN | Test IC 45 | NRMSE (util.py)

T0 = 1181.9 K P0 = 4.842 atm phi0 = 1.099 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	8.22e-03	0.379	1.98	1.07	0.214	0.942	0.0629	1.61	5.76	0	Complete
2 P-only DeepONet10	**10k**	0.25	1.59	0.904	5.59	2.08	2.48	0.364	2	7.4	0	Complete
2 P-only DeepONet10	**long**	0.204	1.41	0.962	5.54	2.02	1.61	0.301	0.138	0.592	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.148	0.982	0.896	1.99	1.19	2.01	0.236	2.46	12.9	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.0104	0.169	0.394	0.29	0.27	0.144	0.0281	0.433	0.46	0	Complete
4 T/P DeepONet10	**10k**	0.137	1.06	1.85	0.835	2.96	1.05	0.199	4.01	5.37	0	Complete
4 T/P DeepONet10	**long**	0.359	0.195	5.55	0.676	0.497	5.43	0.334	5.79	4.53	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.0687	0.168	0.958	0.974	0.537	0.996	0.0649	1.19	0.682	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.302	0.443	3.09	0.817	0.587	5.53	0.334	4.18	6	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.0835	0.76	0.851	1.55	2.22	0.889	0.134	2.86	5.55	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.238	0.193	3.42	0.516	0.299	3.76	0.234	3.01	2.99	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.0594	0.181	0.981	0.969	0.723	0.997	0.0499	0.892	0.616	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	6.98e-03	0.0308	0.0829	0.0481	0.0741	0.134	8.99e-03	0.135	0.296	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.